

Tassadit DEBIANE

Generative AI · Computer Vision · VLM & LLM

+33 7 43 35 98 16 | tassaditdebiane01@hotmail.com | Paris, France

Education

Université Paris Cité

2024 – 2026

M.Sc. Data Science & Artificial Intelligence - High Honors (M1 mention Très Bien)

Paris, France

Experience & Projects

Apprenticeship - Zorba Group

Oct. 2025 – Sept. 2026

Generative AI for Imaging: Conditional Generation, VLMs & Deployment

Paris, France

- **Conditional image generation pipeline:** built an end-to-end pipeline for controllable image synthesis (SDXL + ControlNet, then Flux) conditioned on multi-class segmentation maps, from data ingestion to evaluation by domain experts (1000+ images)
- **Diffusion fine-tuning (LoRA):** LoRA adaptation of Flux on domain-specific data (233 images, kohya-ss); diagnosed and fixed a weight-conversion bug (fused QKV+MLP in Flux blocks); training on dual RTX 5090 GPUs (BFLOAT16, mixed precision, gradient checkpointing)
- **Vision-language model benchmarking:** evaluated Pixtral 12B and Idefics3 8B on captioning; built an evaluation pipeline (BLEU/ROUGE/CIDEr/METEOR) with a human-annotation interface and characterized failure modes
- **Representation & robustness:** characterized the failure of a pretrained vision encoder (UNI2-h) under domain shift via latent-space analysis (in-distribution vs. out-of-distribution)
- **Deployment:** packaged models with Docker and shipped a Gradio application for production use

M2 Research Project - LLM-Augmented Clustering

Jan. – March 2026

Université Paris Cité - Reproduction & Extension of Viswanathan et al., TACL 2024

Paris, France

- Implemented and benchmarked 8 weakly supervised clustering strategies using GPT-4.1-nano (Bank77, CLINC150, Tweet); structured failure-mode analysis across label-noise and sparsity regimes; jury defense and A0 poster

Research Internship - Centre Borelli (CNRS)

April – July 2025

Representation Learning & Unsupervised Evaluation

Paris, France

- **Representation learning:** designed marginalized denoising autoencoders (mDAE) for robust representations on mixed continuous/categorical data; closed-form training objective via analytical marginalization of the corruption distribution
- **Evaluation:** spectral clustering in the learned latent space; rigorous internal/external metrics (Silhouette, Davies-Bouldin, ARI, NMI) against K-means and DBSCAN baselines under label scarcity

Technical Skills

Generative AI: diffusion models (SDXL, Flux), ControlNet, LoRA fine-tuning, conditional generation, failure-mode analysis

Computer Vision & VLM: image generation, semantic segmentation, vision encoders (UNI2-h, DINOv2), VLM evaluation (Pixtral, Idefics3), captioning metrics

NLP / LLM: LLM-augmented pipelines, prompting, LLM-based clustering, evaluation protocol design

Deep Learning: PyTorch (advanced), transformers, VAEs, attention mechanisms, fine-tuning, transfer learning, mixed precision (BFLOAT16), quantization (4-bit NF4)

ML & Representation: representation learning, denoising autoencoders, clustering (spectral, K-means, DBSCAN), self-/semi-/weakly-supervised learning, distribution-shift / OOD analysis

MLOps & Cloud: AWS (SageMaker), Docker, Gradio deployment, Git; dual RTX 5090 (32GB) server, RTX 5070 Ti (16GB)

Languages: Python (expert), R, SQL; French (native), English (proficient, scientific)

Interests

Chess · Fitness, hiking, tennis, badminton · Volunteer at Linkee (food aid)